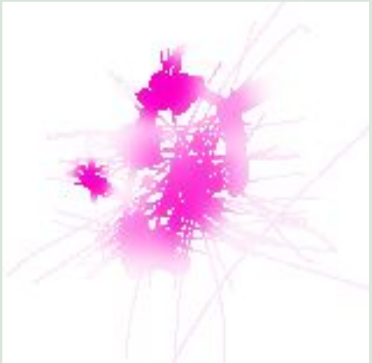
AquaCanvas

Real-Time Watercolor Simulation

A digital painting tool that simulates water flow, pigment transport, and drying over time.



Team Dubuworld

Daeyoung Kim
Jaegyeom Kim
Brian Sui
Shenrui Wu

What We Built So Far

CPU prototype running interactively on a small canvas

Working CPU simulation on a 64×64 canvas

All 5 physics stages run each frame: water displacement, momentum, evaporation, pigment diffusion, and deposition

Improved water transport with a 4-direction redistribution scheme

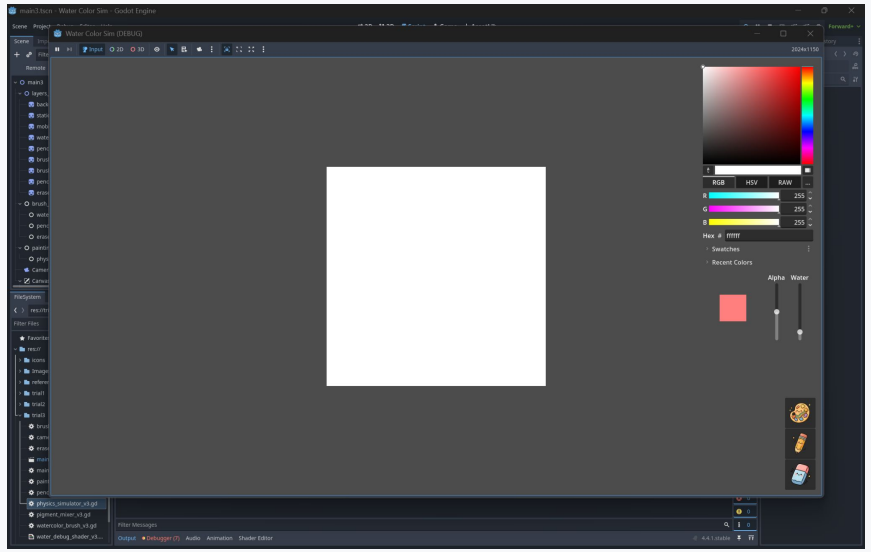
Fixed checkerboard artifacts from the original paper formulation

Built tools: pressure-sensitive watercolor brush, pencil, eraser, color picker, and tilt controls (WASD)

- At 64×64, the CPU version runs at interactive frame rates

Simulation Pipeline

1. Water displacement
2. Inertia / momentum
3. Evaporation
4. Pigment diffusion
5. Deposition



What’s Next

Move from correctness on CPU to scale and performance on GPU

1. Port to GPU

Port the simulation to GLSL compute shaders using Godot RenderingDevice API

2. Validate Output

Make sure GPU output matches CPU results at 64×64

3. Scale Resolution

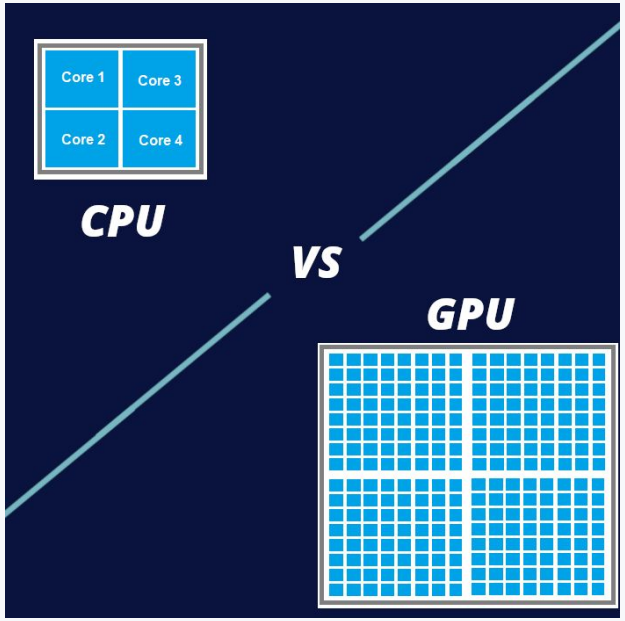
Scale to 512×512 or larger while preserving watercolor behavior

4. Benchmark

Compare CPU vs. GPU FPS and measure the speedup

Main remaining work

Port the CPU prototype to compute shaders for performance

Keep the math and behavior consistent during the CPU → GPU transition

- Show clear evidence that larger canvases become practical with the GPU version